

L2 - 11.3. MISP

Task 1 - Introduction

In this room, you will pivot from a single hash into a full campaign picture using a pre-populated MISP instance, and learn how to integrate MISP into your SOC processes.

Task 2 - MISP Purpose

MISP (Malware Information Sharing Platform) is an open-source threat information platform that facilitates the collection, storage, and distribution of threat intelligence within a community of trusted members. Unlike VirusTotal and similar services, MISP won't give you IP or hash reputation out of the box. Instead, it covers some very specific needs of mature security teams, MSSPs, and government CERTs.

MISP For Small SOC Teams: Small SOC teams dealing with one or two incidents a year won't see the full value of operating a MISP instance, but they can still run one to collect and centralize feeds from threat intelligence producers, and distribute to SIEM, EDR, or IDS platforms. This isn't the ideal use of MISP, but it is how many companies adopt it. To get started with MISP at a minimum, you can:

- Collect events from CERTs and threat feeds, such as CIRCL([opens in new tab](#))
- Push collected indicators (hashes, domains, IPs) from MISP to SOC tools
- Create detection rules that look for indicators in logs or corporate traffic

MISP For Large Teams and CERTs: MISP is the right tool when you need to track the adversaries targeting your organisation, cross-link related activity, and share what you find with partners. Large SOC teams, MSSPs, and government CERTs that handle hundreds of incidents a year get the most out of it, because MISP answers questions like:

Have we seen this indicator before, and in which incident?

Which threat actor or campaign is this activity associated with?

Are multiple organizations being targeted by the same adversary right now?

Has another CERT already observed this artifact? What did they conclude?

What were we dealing with over the past few months? What has changed?

Task 3 - Using MISP

MISP is built around events, collections of linked information describing a single threat, intrusion, or campaign. An event acts as a container that groups everything analysts have observed about one incident. Imagine your SOC just finished responding to a phishing campaign and wants to add the details in MISP. To do this, you'd need to create an event, let's say "Phishing campaign targeting finance staff - May 2026":

The event created will be visible to the organisations having an account on this platform, but not synchronised to other MISP instances until it is published.

Add Event

Date: 2026-05-31 Distribution: This community only

Threat Level: Undefined Analysis: Initial

Event Info

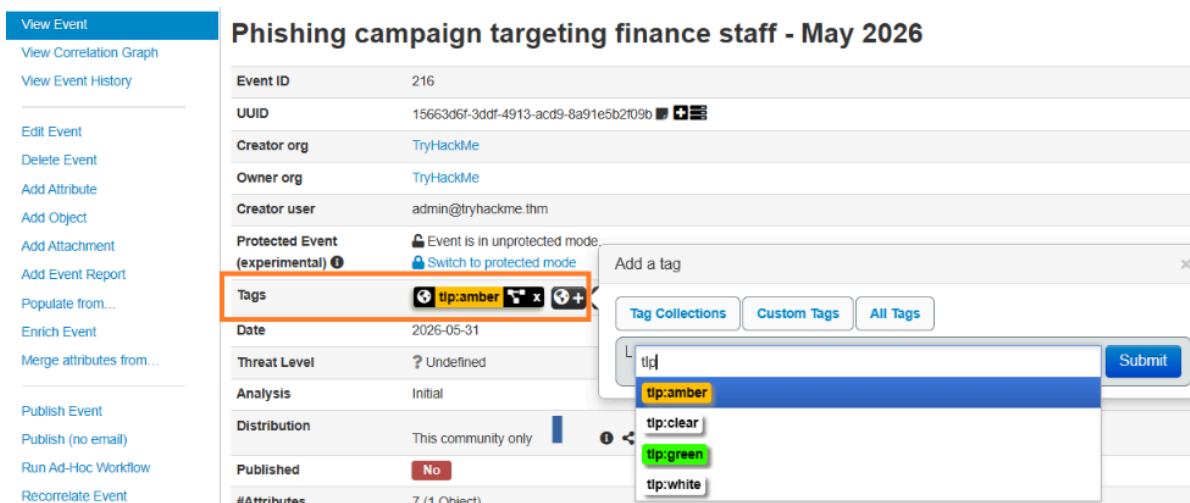
Phishing campaign targeting finance staff - May 2026

Extends Event

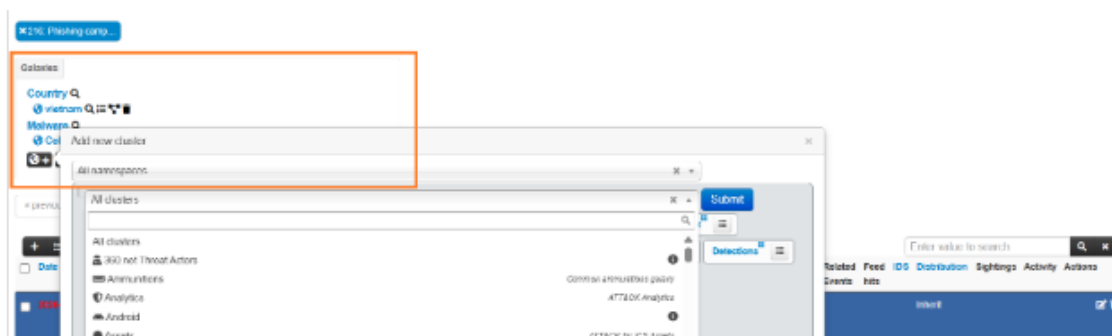
Event UUID or ID. Leave blank if not applicable.

Submit

Tags and TLP: Before filling in the event, you need to decide its sharing rules. In MISP, you can do it with TLP classification and event tags, where tlp:clear tag would mean the event can be publicly shared, and tlp:amber means the event is meant only for internal use and trusted partners. We will come back to tags and TLP in the next task, but for now, follow the screenshot and assign a single event tag:



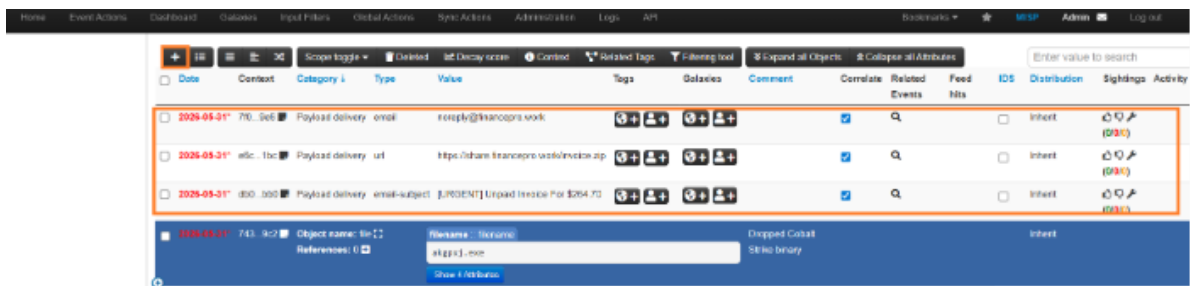
MISP Galaxies: Let's say you know the phishing attack originates from Vietnam, and ends up with Cobalt Strike malware being installed on a victim's PC. To mention it in the event, use MISP Galaxies - libraries of structured knowledge with a list of countries, tools, malware families, and many more. The more Galaxies you add, the easier it is to group and link the events. Follow the screenshot below and add two Galaxies: one for the origin country, and one for the used malware:



Attributes: Finally, you can fill in the event with attributes, such as IPs, domains, hashes, and free-text comments. A good event has as many attributes as possible, and an ideal event also has text comments, references, or even exported phishing emails and malware binaries inside. Let's add the following:

The sender address noreply@financepro[.]work (attribute: email-src)

The phishing URL [https://share.financepro\[.\]work/invoice.zip](https://share.financepro[.]work/invoice.zip) (attribute: url)



Objects: MISP objects are containers for related attributes. For example, we could have grouped sender address, email subject, and phishing URL into the email object. Same for the file object that can group file name, hash, size, and path. It is preferred to use objects instead of independent attributes, so let's create a file object for the dropped Cobalt Strike malware with some common attributes (Event > Left Panel > Add Object > File):

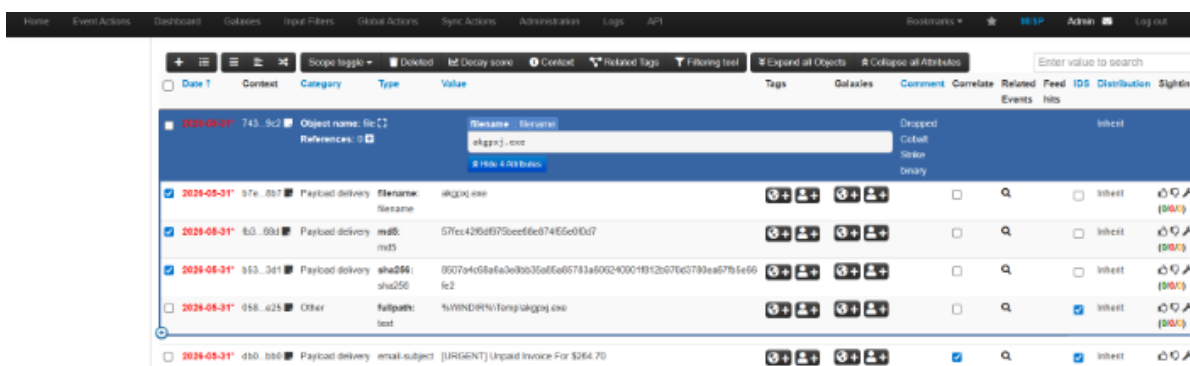
The filename of the observed malware: akgpxj.exe (attribute: filename)

It's absolute file path: %WINDIR%\Temp\akgpxj.exe (attribute: fullpath)

It's MD5 hash: 57fec42f6df975bee68e874f55ef0fd7 (attribute: md5)

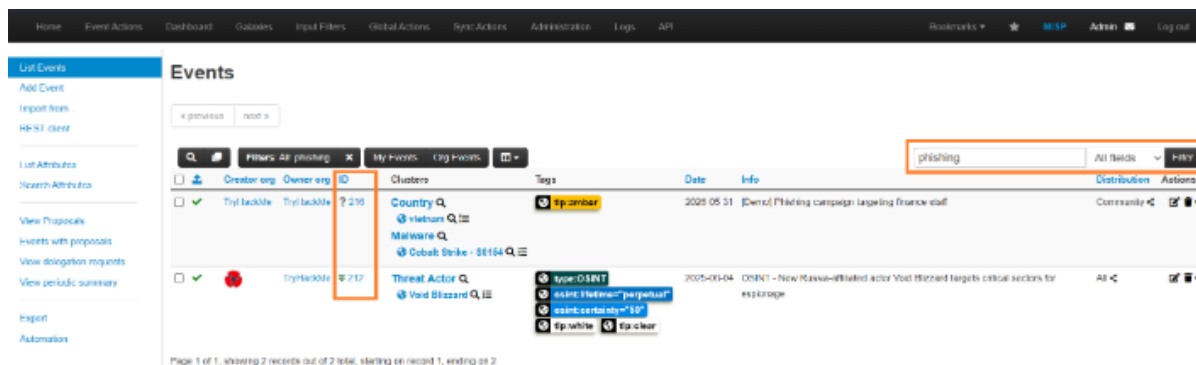
It's SHA256 hash:

8507a4c68a6a3e8bb35a85783a606240901f812b970d3780ea67fb5e66fe2
(attribute: sha256)



What's Next: Once analysts have created an event, a user with the publish permission reviews and publishes it (Event > Left Panel > Publish). Publishing makes the event visible according to its distribution level, and is pushed to sync partners if any are configured. Then, you can link the event back to your SOC ticketing system using its UUID. The next time you or your partners face an incident, the published event can help with the investigation. You can:

Search prior events by attribute (hash, IP, domain, etc.) or free-text
 Attribute the activity to a known threat actor via galaxies or correlated tags
 Accelerate response with the linked TTPs, infrastructure, and analyst notes



SOC ticketing systems track work in progress, but MISP accumulates reusable knowledge about threats and APT groups.

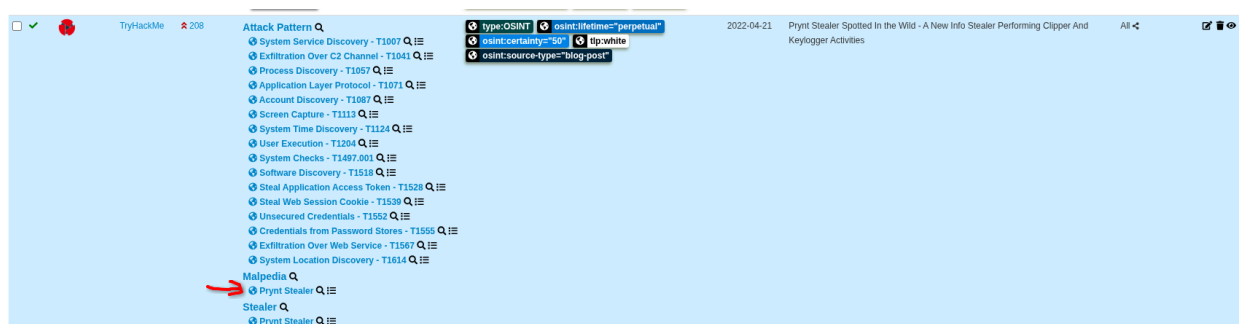
Answer the questions below

How do you call an atomic indicator in MISP, such as IP address?

Attribute

What malware name is related to the following attribute?
 MD5 Hash: 661842995f7fdd2e61667dbc2f019ff3

Prynt Stealer



Task 4 - Feeds and Taxonomies

Taxonomies: A taxonomy is a means of classifying information based on standard features or attributes. On MISP, taxonomies are expressed in event tags, which

comprise three vital parts: Namespace (Defines the tag's property to be used). Predicate (Specifies the property attached to the data). Value (Numerical or text details to map the property).

Taxonomy Examples:

tlp:red: Do not share outside the meeting

type:OSINT: The event is open-source intelligence

retention:30d: The event should expire after 30 days

admiralty-scale:source-reliability="a": Completely reliable source

The screenshot shows the 'Taxonomy Tags' interface in MISP. It includes a search bar, a filter dropdown set to 'All', and a table of tags. The table has columns for Name, Expanded, Numerical Value, # Events, # Attributes, Tag, Enabled, and Actions. Three tags are listed: 'information-orig:AI-generated', 'information-orig:human-generated', and 'information-orig:unknown-origin'. Each tag has a value of 0 for both # Events and # Attributes, and is currently disabled.

Name	Expanded	Numerical Value	# Events	# Attributes	Tag	Enabled	Actions
information-orig:AI-generated	AI generated	0	0			Not	
information-orig:human-generated	human generated	0	0			Not	
information-orig:unknown-origin	unknown origin	0	0			Not	

Page 1 of 1, showing 1 records out of 3 total, starting on record 1, ending on 3

One of the prebuilt taxonomies, showing whether the report/event was AI-generated

(Top Menu > Event Actions > List Taxonomies)

MISP Feeds: You've learned how to create and search for events. Now, let's see how to import the events of other organizations using feeds. By default, the MISP instance comes with two public feeds: Computer Incident Response Center Luxembourg (CIRCL) and botvrij.eu(opens in new tab) initiative. You enable both of them, and then decide whether you want to cache or fetch the events:

Fetch: Imports full events into your MISP instance, useful for sync between trusted partners

Cache: Saves indicators in a local cache for correlation, but does not import full events

(Cache is useful when you don't want to pollute your instance with events from noisy public feeds like CIRCL)

MISP Communities: MISP delivers its highest value when used within an active sharing community. Most communities are private and membership-gated, but if your organization operates in a regulated sector, critical infrastructure, or government, you may qualify to join one and gain access to high-quality intelligence. A few examples of MISP communities:

NATO Cyber Security Center

Forum of Incident Response and Security Teams (FIRST)

National CERTs, such as CERT-EU, CERT-UA, and many others

Task 5 - Scenario Event

Scenario: CIRCL (Computer Incident Response Center Luxembourg) republished a CERT-UA event associated with an APT28 attack on European organizations. The adversary exploited a critical Microsoft Office CVE to hit major European entities, and your company is concerned it might be next. You have been tasked with investigating this campaign and correlating the details with your SIEM. Use what you have learned in this room to identify the event details and complete this task. Good luck!

Answer the questions below

What event ID has been assigned to the APT28 event?

211

Refer to the event galaxies.

How does Microsoft refer to APT28 group?

STRONTIUM

What CVE, targeting MS Office, was used in the attack?

CVE-2026-21509

What is the MD5 hash of the dropped Covenant C2 DLL?

6f528ad405bffa4a8c2f61b1fa2172fd

md5 hash created by covenant dll

2026-02-02	706...b62	Object name: file	filename :: filename	filename :: filename
		References: 0		covenant.dll
			Hide 3 Attributes	
☐	2026-02-02	d07...26e	Payload delivery	filename: covenant.dll
			filename	
☐	2026-02-02	721...5f0	Payload delivery	md5: 6f528ad405bffa4a8c2f61b1fa2172fc
			md5	
☐	2026-02-02	22e...dd4	Payload delivery	sha256: 40c2e559992a7f595c593b419930a3f216516c3042ad86fb985348d53b6e01b9
			sha256	

What is the URL of the CERT-UA report referenced in the event attributes?

<https://cert.gov.ua/article/6287250>

Last url in the event:

2026-02-02	090...dda	Object name: report	link :: link	link :: link
		References: 0		https://cert.gov.ua/article/6287250
			Hide 3 Attributes	
☐	2026-02-02	eaf...4a5	External analysis	link: https://cert.gov.ua/article/6287250
			link	
☐	2026-02-02	c5b...8f8	Other	title: "Бюлетень безпеки": UAC-0001 (APT28) здійснює кібератаки у відношенні України та країн ЄС з використанням експлойту CVE-2026-21509 (CERT-UA#19542)
			text	
☐	2026-02-02	09d...a84	Other	title: "Hanger Bulletin": UAC-0001 (APT28) carries out cyberattacks against Ukraine and EU countries using the exploit CVE-2026-21509 (CERT-UA#19542)
			text	

Task 6 - Conclusion

MISP use in sharing malware and threat information in this room. This tool is useful in the real world regarding incident reporting. You should be able to use the knowledge gained to effectively document, report and share incident information.

Strontium (more commonly known as **APT28**, **Fancy Bear**, or **Sofacy**) is a well-known, state-sponsored cyber espionage group.

- **Attribution:** Security researchers and governments widely attribute Strontium to the **GRU** (Russia's military intelligence agency), specifically operating under units like Unit 26165.
- **Targeting:** The group primarily targets government agencies, military bodies, defense contractors, international think tanks, political organizations, media outlets, and critical infrastructure—particularly in NATO countries, the U.S., and Europe (with a heavy focus on Ukraine).
- **Notable Operations:** Strontium has been linked to major global cyberattacks, including the 2016 Democratic National Committee (DNC) hack, the 2015 intrusion into the German Parliament (Bundestag), and ongoing espionage campaigns tracking geopolitical developments and the conflict in Ukraine.
- **Tactics:** They commonly employ spear-phishing campaigns, credential harvesting, zero-day exploits, custom malware payloads (such as XAgent and Zebrocy), and the exploitation of weak IoT device configurations to gain initial access to corporate networks.

Prynt Stealer (often misspelled or referred to phonetically as **Print Stealer**) is a type of **information-stealing malware** (infostealer) sold on cybercrime forums under a Malware-as-a-Service (MaaS) model. First discovered around 2022, it is designed to harvest sensitive data from infected Windows systems and facilitate direct financial theft.

Key Capabilities

- **Data Harvesting:** Targets dozens of Chromium- and Firefox-based web browsers, extracting saved passwords, autofill data, credit card details, and session cookies. It also steals data from VPN clients, FTP software, messaging apps (like Telegram and Discord), and gaming platforms.
- **Clipper Module:** Features a built-in "clipper" that monitors the system clipboard. If it detects a user copying and pasting cryptocurrency wallet addresses, it automatically replaces the destination address with the attacker's address to redirect funds.
- **Keylogger:** Includes native keylogging functionality to record keystrokes, allowing threat actors to capture sensitive text, chat messages, or newly typed credentials in real time.

- **Builder Customization:** Sold via subscription models (ranging from monthly to lifetime licenses), it provides threat actors with an easy-to-use builder to compile customized, evasion-optimized builds of the malware.
- **Exfiltration via Telegram:** Often utilizes Telegram bots or channels as a lightweight command-and-control (C2) mechanism to ship harvested logs back to the attacker.